

S.M. MAHBUBUL HASAN

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Present Address: Basundhara R/A, Dhaka, Bangladesh | Permanent Address: Jubli Bagan Len, Sirajganj, Bangladesh

CAREER OBJECTIVE

CSE graduate with hands-on experience in Python, Deep Learning, and Computer Vision. Built and deployed real-world AI solutions including a YOLOv8-powered waste classification system (mAP@0.5: 0.919) and a SegFormer-based brain tumor segmentation model. Proficient in developing end-to-end ML pipelines and integrating AI models into web applications. Seeking an entry-level role to contribute to scalable and impactful AI products while growing as an engineer.

EDUCATION

Bachelor of Science in Computer Science and Engineering | December 2025

North South University, Bashundhara, Dhaka-1229, Bangladesh

- Relevant Coursework: Machine Learning, Artificial Intelligence, Theory of Computation

RELEVANT COURSEWORK

CSE 445 - Machine Learning (2025) Supervised and unsupervised learning, neural networks, clustering, decision trees, evolutionary algorithms

CSE 440 - Artificial Intelligence (2025) Search, knowledge representation, NLP, computer vision, robotics; LISP and PROLOG

CSE 273 - Theory of Computation (2024) Propositional logic, finite automata, formal languages, Turing machines, NP-completeness

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript

Web Technologies: HTML, CSS, JavaScript

Machine Learning and AI: Deep Learning, Computer Vision, YOLOv8, SegFormer, EfficientNetB4

Frameworks and Tools: TensorFlow, PyTorch, OpenCV, Streamlit, Git, GitHub, Roboflow, Google Colab

Core Concepts: Data Preprocessing and Augmentation, Model Training and Evaluation, Image Classification and Segmentation

Soft Skills: Problem Solving, Teamwork, Communication, Quick Learning, Time Management

PROJECTS

Real-Time Waste Classification System Using YOLOv8 | 2025

Senior Design Project | North South University | Python, PyTorch, YOLOv8, OpenCV, Streamlit, Roboflow

- Trained YOLOv8s on 11,709 images across 23 waste categories and achieved mAP@0.5 of 0.919, Precision of 0.904, and Recall of 0.882.
- Deployed real-time web application using Streamlit and OpenCV with live webcam inference, batch image processing, and automated waste sorting simulation.
- Applied data augmentation, normalization, early stopping, dropout, and weight decay to maximize model generalization.

Brain Tumor Segmentation Using SegFormer | 2025

Directed Research | North South University | Python, PyTorch, SegFormer, OpenCV, Google Colab

- Developed a deep learning model using SegFormer for pixel-level brain tumor segmentation in MRI images.
- Applied preprocessing pipeline including resizing, normalization, and augmentation to improve model robustness.
- Trained and evaluated the model on standard segmentation metrics for accurate tumor boundary detection.

EXTRACURRICULAR ACTIVITIES

- Core Member, NSU Art and Photography Club: contributed to event management and creative initiatives.
- General Member, NSU ACM Student Chapter.
- Participated in university tech events, workshops, and collaborative project activities.

LANGUAGES

Bengali: Native

English: Professional Proficiency (Reading, Writing, Speaking)

REFERENCES

Dr. Sifat Momen

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Md. Moznu Sarkar

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